UAE HACKATHON 2026 · CHALLENGE #27 · EVIDENCE AND DATA USE

Evidence brief وكالة

Every number in the pitch, where it comes from, and what kind of data it is.

SYNTHETIC PROTOTYPE BENCHMARK · NOT PRODUCTION GOVERNMENT STATISTICS

1. Data sources at a glance

Source	Type	Used for	Location
Challenge #27 profile (Cyber Security Council, organiser-provided)	**Official challenge material**	Problem framing: the national shift of half of government services and operations to agentic AI within two years, and root causes R1–R5	Provided by the organisers (not redistributed in this pack)
HackArena Playbook 2026, Track 2	**Official hackathon material**	Format (3-minute pitch + 2 minutes of questions), judging criteria, 90-day plan requirement	Provided by the organisers
Fixed-seed labelled dataset: 1,200 requests in 20 categories	**Synthetic** (generated by WAKALA)	Security benchmark	`evidence/phase5/dataset.json`, `summary.json`, `REPORT.md`
Recovery regression and repeatability runs	**Synthetic**	Behaviour Watch before/after, repeatability	`evidence/phase6/`, `evidence/phase7/recovery_repeatability.json`
Demo run (citizens C-1001 to C-1003, fictional authorities, synthetic treasury)	**Synthetic**	Screenshots and the live demo	`submission/screenshots/CAPTURE_LOG.json`
Hypothetical efficiency calculator	**Assumptions only**	Not used in the pitch	`evidence/phase5/REPORT.md` §8

No official government data and no personal data were used. WAKALA authorizes and records the actions agents take. It does not need real citizen records to show that an out-of-mandate action is blocked. We used synthetic records instead of real ones, following the playbook's rule against uploading personal or unpublished government data. Fictional entities, synthetic citizens and a synthetic treasury are labelled as such on every screen. No UAE open dataset feeds the benchmark. In a pilot, the entity's own test data would replace the synthetic records (see the 90-day roadmap).

2. Claim → figure → source

Claim in the pitch	Figure	Exact source
Unauthorized operations executed under full WAKALA	**0 / 720**	`evidence/phase5/summary.json` → configuration C, `unauthorized_operations_executed`
Legitimate requests incorrectly denied under full WAKALA	**0 / 480**	same file, C, `legitimate_incorrectly_denied`

Claim in the pitch	Figure	Exact source
Intentionally weak role-only baseline executed	**624 / 720** (86.7 %)	same file, configuration A
Case-bound WAKALA without delegation (B)	0 / 575 executed; 41 / 417 legitimate denied (officer scope, explained in the report)	`REPORT.md` §3
Attack Lab	Baseline executed **7 / 8**; WAKALA **0 / 8**; attack B denied by both	`docs/PHASE6.md` M2; `docs/FINAL_TECHNICAL_READINESS.md` §5 step 8
Shared-environment recovery, original policy	**2 / 20** (historical and reproduced in Phase 7)	`evidence/phase5/REPORT.md` §7; `FINAL_TECHNICAL_READINESS.md` §4
Shared-environment recovery, Behaviour Watch 2.0	**20 / 20** in each of 5 repeats; **50 / 50** long run	`evidence/phase7/recovery_repeatability.json`
Every attack still contained in the recovery runs	20 / 20 denied and not executed per run (2 suspensions, 18 revocations)	same file
Gateway latency	≈ 19 ms median full authorization in memory; ≈ 55 ms with persisted SQLite	`REPORT.md` §5 (direct 18.68 / 54.80 ms; delegated 19.16 / 55.77 ms)
Test suite	**493** backend tests pass, including 34 security invariants	`FINAL_TECHNICAL_READINESS.md` §1, §3
Demo verified end-to-end	**12 / 12** steps, 15 routes render	`FINAL_TECHNICAL_READINESS.md` §1, §5
Shadow Mode prediction vs enforcement	**11 / 11** matched	`docs/PHASE6.md` M5
Demo audit chain	Verified, 386 events at capture	`submission/screenshots/CAPTURE_LOG.json`

3. How to read these numbers

- **Synthetic, deterministic, single machine.** The benchmark ran on one Windows 11 laptop (Ryzen 7 5700U) in one Python process with SQLite. The results show the logic is correct and reproducible. They are **not** production throughput or reliability figures.
- **The baseline is deliberately weak** (it checks the role only). It is a lower bound, not a model of existing UAE government IAM. Case-bound configuration B is the fairer comparison.
- **Labels come from request categories,** never from any configuration's decision. Seed `20260`; the dataset SHA-256 is recorded in `manifest.json`.
- **Prevention is not the same as detection.** Monitoring rules flagged 657 / 720 unauthorized requests. Expired-mandate requests, for example, are prevented by authorization but have no dedicated monitoring rule.
- **Recovery fix.** The 2/20 result was a real weakness: agent-wide counters attached incidents to unrelated cases. It was diagnosed and fixed, and the historical result is kept and shown.

4. What is not evidenced

- **No live AI model was tested.** The demo uses a labelled, deterministic offline planner. Nothing is claimed about how a real model behaves with injected text.
- **No integration with any government system.** No real payments or real citizen records.
- **No production deployment, high availability or external audit anchoring.**

- Behaviour Watch thresholds are hand-set rules, not AI threat prediction.

5. Reproduce

```
cd backend
.venv\Scripts\python -m app.evaluation.recovery_repeat
```

The full Phase 5 benchmark takes about 30 minutes: `.venv\Scripts\python -m app.evaluation.run --seed 20260 --size 1200 --out ../evidence/phase5`
